

CELLINSIGHT

AI-Assisted Digital Peripheral Blood Smear Review Platform

Complete Product Requirements Document (PRD)

Implementation-ready specification for the final-year B.Tech project

Project	CellInsight: AI-Assisted Blood Smear Analysis System
Domain	AI + Data Science + Computer Vision + Web Engineering
Target setting	Small pathology / diagnostic laboratory workflow
Development window	8 weeks / approximately 2 months
Core AI retained	YOLOv8 cell detector + EfficientNet/Keras WBC classifier
Primary users	Laboratory technician, doctor/pathologist, laboratory administrator

What is changing?

Previous: upload one smear image → run YOLO/EfficientNet → display counts → download PDF. **New:** laboratory case management → image quality check → multi-image analysis → AI detection/classification → confidence and review prioritisation → human verification → digital record → approved report. The AI models remain the core inference engine; the project is expanded into an end-to-end laboratory assistance system.

Prepared as the implementation blueprint for the student development team and coding agent.

0. Document Control

Field	Value
Document	CellInsight Complete Product Requirements Document
Version	1.0 — Implementation Baseline
Status	Build-ready specification
Project title	CellInsight: AI-Assisted Blood Smear Analysis System
Academic year	2026–27
Scope horizon	8 weeks
Primary constraint	No dedicated project budget; prefer free/open-source tools and existing hardware
Clinical positioning	Research/educational assistance prototype; not a diagnostic or autonomous medical decision system

Source basis

- Review-1 project review document supplied by the team.
- Current project synopsis supplied by the team.
- Shared implementation snapshot describing the current three-service codebase.
- External research and product references used only to strengthen the workflow and engineering rationale.

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1. Executive Summary

CellInsight is being redesigned from a narrow blood-smear image analysis demo into a low-cost, human-in-the-loop digital workflow for peripheral blood-smear review in a small laboratory setting. The project retains the existing AI foundation—YOLO-based detection of RBCs, WBCs and platelets and a Keras/EfficientNet WBC classifier—but places those models inside a complete case-management and review system.

The system is not intended to replace CBC analysers, laboratory professionals, doctors, or existing diagnostic reports. Its purpose is to reduce repetitive manual work around smear-image review: organising cases, checking image suitability, pre-classifying cells, prioritising uncertain cells, recording reviewer corrections, preserving a digital history, and generating a structured report after human verification.

This direction is consistent with established digital morphology workflows in which AI performs cell pre-classification and a medical technologist reviews and verifies the result. The ICSH review also identifies digital archiving, quality assurance, consultation, education and workflow standardisation as useful functions around digital morphology, while emphasising the continuing role of skilled morphologists in reclassification and interpretation. [External research: ICSH 2019; CellaVision product documentation.]

Product statement

CellInsight helps a laboratory turn peripheral blood-smear images into organised, reviewable, traceable analysis records, with AI assisting rather than replacing human verification.

Primary value proposition

- One place to manage smear cases and previous results.
- Reject or warn on unsuitable images before AI analysis.
- Use existing AI models as an assistant instead of presenting raw model output as a final diagnosis.
- Send low-confidence or abnormal-looking cells into a focused review queue.
- Preserve AI predictions and reviewer corrections for traceability and future model improvement.
- Measure not only model metrics but also practical workflow metrics such as review time and correction rate.

2. Current Project Baseline

The supplied implementation snapshot describes a small three-service application: Python/FastAPI inference, Node/Express backend, and React/Vite frontend. The current AI pipeline is real and should be preserved rather than discarded.

Layer	Current implementation	Keep / Change
Inference	YOLO model detects WBC, RBC and platelets; WBC crops are classified by Keras/EfficientNet.	KEEP as core AI engine; improve outputs and integration.
Backend	Express receives image, forwards to inference service and returns results.	KEEP; expand into case, user, review and report APIs.
Frontend	Image upload, health indicator, result counts, WBC subtype list and PDF export.	KEEP UI foundation; expand into lab dashboard and review workspace.
Decision logic	Simple placeholder: WBC count > 20 → review suggested; otherwise normal range.	REMOVE as a clinical risk rule; replace with confidence/review prioritisation.
PDF	Styled PDF references blast percentage and reliability note that are not fully produced by backend.	REBUILD as verified-case report; implement fields only when actually computed.
Database	Synopsis mentions MongoDB, but the shared current implementation snapshot does not establish a complete case/history database.	ADD as a first-class module.

Current model classes

- YOLO: WBC, RBC, Platelets.
- Current WBC classifier: basophil, erythroblast, monocyte, myeloblast, segmented neutrophil.
- **Important taxonomy task:** erythroblast/NRBC should not be treated as a normal WBC subtype in the final system. The final taxonomy must be reviewed and fixed before retraining or evaluation.

3. Problem, Gap and Why We Are Changing the Scope

The original project statement is retained because it has already been submitted and reviewed. The scope around that statement is being improved.

Original problem framing

The supplied synopsis states that manual blood-smear analysis is time-consuming, tedious, requires specialised personnel, and can vary when large numbers of samples are examined; it proposes automated detection and counting from microscopic images. The review presentation similarly positions AI as a way to reduce manual effort and improve consistency.

The panel's practical challenge

A single-image-to-PDF application does not clearly explain why a laboratory would adopt it when laboratories already have established reporting processes. The redesign therefore moves the value from PDF generation to the workflow before the report: case management, image quality, AI pre-analysis, prioritised human review, traceability and quality analytics.

External benchmark and gap

- Digital morphology systems can pre-classify cells and then have a medical technologist review and verify the classifications.
- They can also support comments, comparison with previous results, collaboration, consultation and archival.
- The proposed student system cannot reproduce the full capability of a commercial analyser, but it can implement a smaller, low-cost workflow around existing microscope images and open-source software.
- The engineering gap for this project is therefore **integration and workflow intelligence**, not invention of a new neural-network architecture.

Project-level engineering contribution

Build an end-to-end, low-cost, human-in-the-loop workflow in which existing AI models are made useful through image-quality gating, confidence-aware review prioritisation, human verification, digital case history, auditability and measurable workflow evaluation.

4. Product Vision and Goals

Vision

Create a practical laboratory-side web application that helps staff organise peripheral blood-smear cases, use AI to pre-analyse microscopic images, review uncertain results efficiently, and preserve verified results as digital records.

Goals

- Provide a complete case-based workflow rather than isolated image uploads.
- Support one or more smear images per case.
- Introduce an image-quality gate before AI inference.
- Retain and integrate the existing YOLO and EfficientNet models.
- Add confidence-aware review prioritisation.
- Allow a reviewer to accept, change or mark AI classifications as uncertain.
- Store both AI output and final reviewer output.
- Generate a structured report from verified results.
- Provide a laboratory dashboard for pending, processing, review and completed cases.
- Evaluate both model performance and real workflow performance.

Engineering principles

- **Human-in-the-loop:** AI assists; authorised human users verify.
- **Traceability:** preserve what the AI predicted and what the reviewer changed.
- **Uncertainty-aware:** do not force low-confidence predictions into a final answer.
- **Modular:** AI service, API, database and frontend remain independently testable.
- **Feasible:** use free/open-source tools and existing hardware wherever possible.
- **Safe positioning:** prototype for research/education; no autonomous diagnosis.

5. Users and Roles

Role	Main responsibilities	Permissions
Laboratory Technician	Create cases, add images, run analysis, inspect AI results, perform first-level verification.	Create/update cases; upload images; view AI; review cells; submit for doctor review.
Doctor / Pathologist	Review flagged/uncertain cells, make final corrections, approve report.	All review permissions; final verification; approve/sign report.
Laboratory Administrator	Manage users, view workload, monitor performance and records.	User management; dashboard; archive access; analytics.
Coding/ML Team	Develop, test and evaluate the system.	Development-only access; no production patient workflow.

Out of scope: patient portal

The project will remain on the clinical/laboratory side. Patients will not have a separate portal. Approved reports may be shared by the laboratory through a controlled mechanism if implemented, but patient-facing account management is not part of the core scope.

6. Scope and Non-Goals

In scope	Out of scope for the 8-week build
Case and sample management	Full hospital information system
Multiple smear-image upload per case	Automated microscope hardware/control
Image quality assessment	Autonomous slide scanning
YOLO cell detection	Replacement of CBC analysers
WBC classification	Final medical diagnosis
Confidence/review prioritisation	Treatment recommendations
Human verification	Patient portal
Digital archive and audit trail	Full LIS/HIS integration
Structured PDF report	WhatsApp API as a core dependency
Model and workflow evaluation	Large-scale cloud deployment

Definition of success

The project succeeds if a laboratory user can create a case, upload multiple images, receive quality feedback, run AI analysis, review flagged/uncertain cells, save verified results, retrieve the case later, and generate a report—all through one coherent application.

7. Functional Requirements

ID	Requirement	Priority
FR-01	User login and role-based access for technician, doctor/pathologist and administrator.	Must
FR-02	Create, edit and view laboratory cases with a unique case/sample identifier.	Must
FR-03	Associate one or more smear images with a case.	Must
FR-04	Run image-quality checks before inference.	Must
FR-05	Allow poor-quality images to be rejected or marked for recapture/re-upload.	Must
FR-06	Run YOLO cell detection on accepted images.	Must
FR-07	Classify detected WBCs using the current classifier or revised validated model.	Must
FR-08	Return cell-level confidence and location information.	Must
FR-09	Generate review-priority flags for uncertain/abnormal-looking cells.	Must
FR-10	Allow authorised users to accept, reclassify or mark a cell as unknown.	Must
FR-11	Store original AI prediction and final reviewer decision.	Must
FR-12	Provide case-level counts and verified summaries.	Must
FR-13	Generate a structured PDF report from verified results.	Must
FR-14	Search and retrieve previous cases and reports.	Must
FR-15	Provide basic dashboard statistics.	Should
FR-16	Track processing/review time and AI-human agreement.	Should
FR-17	Email/share approved report.	Should
FR-18	Camera capture mode.	Could

8. Five Implementation Modules

These are **implementation modules**—how the team will build the project—not merely the runtime workflow.

MODULE 1	Data Preparation & Model Development
Purpose	Prepare datasets, fix taxonomy, train/fine-tune and evaluate the AI components.
Implementation	Organise datasets; preprocess/augment images; validate labels; train/fine-tune YOLO detector and WBC classifier; generate metrics; freeze versioned model files.
Technology	Python, OpenCV, Ultralytics YOLO, TensorFlow/Keras, scikit-learn, Google Colab.
Outcome	Validated model artefacts with documented class labels, metrics and inference interfaces.
MODULE 2	AI Inference & Image Intelligence Service
Purpose	Turn the trained models into a reliable Python service used by the application.
Implementation	Implement image-quality checks; load models once; run detection; crop WBCs; classify; compute confidence and review-priority fields; return structured JSON; expose health/predict endpoints.
Technology	FastAPI, Python, OpenCV, NumPy, TensorFlow/Keras, Ultralytics, Uvicorn.
Outcome	One reusable AI service that the backend can call for every accepted image.
MODULE 3	Application Backend, Database & Security
Purpose	Provide the business logic, data storage and controlled access layer.
Implementation	Implement authentication; role management; case APIs; image metadata; analysis storage; review actions; audit log; report records; validation; error handling.
Technology	Node.js, Express, MongoDB, JWT, bcrypt, Multer, REST APIs.
Outcome	A persistent laboratory system in which every case, image, prediction, correction and report is traceable.
MODULE 4	Laboratory Web Application
Purpose	Give laboratory users a usable interface for the entire process.
Implementation	Build login, dashboard, case creation, image upload, quality feedback, analysis view, cell review gallery, correction tools, case history and report screens.
Technology	React + Vite, JavaScript/TypeScript if adopted, HTML/CSS, Chart.js, jsPDF.
Outcome	A complete browser-based laboratory interface rather than a single upload page.
MODULE 5	Testing, Evaluation, Deployment & Documentation
Purpose	Prove that the system works and prepare it for demonstration.
Implementation	Unit/integration/UI tests; model evaluation; AI-human agreement; time measurements; error/edge cases; Docker/local deployment; user guide; technical documentation; final demo dataset.
Technology	Postman, pytest, browser testing, Git/GitHub, Docker optional, Python evaluation scripts.
Outcome	A reproducible, demonstrable final-year project with measurable results.

9. AI / ML Design

The project does not require invention of a new neural-network architecture. The engineering contribution is the design of an intelligent pipeline around proven models and the measurable use of their outputs.

9.1 Existing AI models to retain

Component	Role	Current baseline	Final treatment
YOLO detector	Detect/localise RBC, WBC, platelets	Existing trained model	Retain, validate and integrate.
EfficientNet/Keras classifier	Classify WBC crops	Existing fine-tuned model	Retain where performance is acceptable; revise taxonomy/training if needed.

9.2 New project-specific intelligence

Logic	Implementation	Purpose
Image Quality Score	OpenCV-based focus/blur, brightness, contrast, resolution and optional density checks.	Stop clearly unsuitable images before expensive inference.
Confidence Handling	Use detector/classifier confidence and probability margin.	Avoid presenting weak predictions as certain.
Review Priority	Combine confidence, image-quality penalty and abnormal/uncertain class flag.	Create a focused review queue.
Human Correction Capture	Store AI prediction + reviewer result + timestamp + user.	Build traceability and future retraining data.
Case-Level Aggregation	Aggregate results across accepted fields/images.	Avoid treating one microscopic field as the entire case.

9.3 Example review-priority logic

This is an implementation starting point, not a clinical threshold. The weights must be tuned and documented during evaluation.

```
review_priority = 0.40 × (1 – classification_confidence) + 0.25 × (1 – detection_confidence) + 0.20 ×
image_quality_penalty + 0.15 × class_ambiguity
```

Suggested product behaviour: low priority → normal display; medium priority → review recommended; high priority → mandatory review. The system must never translate this score into a disease diagnosis.

10. System Architecture

The architecture preserves the current three-tier pattern but expands it into a complete laboratory platform.



Layer	Responsibilities
React frontend	Login, dashboard, case management, image upload, quality feedback, results, cell review, reports, history.
Express backend	Authentication, authorisation, case APIs, image handling, business rules, database access, audit logs, report metadata.
FastAPI AI service	Quality checks, YOLO inference, WBC classification, confidence calculation, review-priority output.
MongoDB	Users, cases, samples, images metadata, AI results, reviewer actions, reports, audit events.
File/object storage	Original images, annotated images, cell crops, generated reports. Local filesystem is acceptable for prototype.

Core data flow

Browser → Express API → FastAPI AI service → Express stores results → Browser displays AI results → reviewer corrections → Express stores verified result → report service generates PDF → report metadata is archived.

11. Data Model

MongoDB is suitable for the prototype because cases naturally contain nested image and analysis information. Keep the schema explicit even if collections are document-based.

Collection	Key fields
users	_id, name, email, password_hash, role, status, created_at
patients/case_subjects	_id, subject_code, optional demographic fields, created_at
cases	_id, case_id, subject_id, sample_id, status, created_by, assigned_to, created_at, updated_at
images	_id, case_id, file_path, field_no, capture_time, quality_score, quality_status, quality_reasons
analyses	_id, case_id, image_id, model_version, started_at, completed_at, cell_counts, status
cells	_id, analysis_id, image_id, cell_type, bbox, detection_confidence, subtype, subtype_confidence, review_priority
reviews	_id, cell_id, original_prediction, final_label, action, reviewer_id, comment, reviewed_at
reports	_id, case_id, version, status, file_path, approved_by, approved_at, created_at
audit_events	_id, user_id, case_id, action, entity_type, entity_id, timestamp, metadata

Data principle

Do not overwrite the original AI prediction when a reviewer changes it. Store the original prediction and the final verified label separately so the system can measure AI-human agreement and preserve an audit trail.

12. API Requirements

Method	Endpoint	Purpose
POST	/api/auth/login	Authenticate user and issue session/JWT.
GET	/api/health	Backend health check.
POST	/api/cases	Create a case.
GET	/api/cases	List/search cases.
GET	/api/cases/:id	Get case details/history.
POST	/api/cases/:id/images	Upload one or more images.
POST	/api/images/:id/quality-check	Run or retrieve image quality assessment.
POST	/api/cases/:id/analyze	Submit accepted images for AI analysis.
GET	/api/analyses/:id	Retrieve analysis and cell results.
PATCH	/api/cells/:id/review	Accept/reclassify/mark unknown; record comment.
POST	/api/cases/:id/approve	Approve verified case/report.
POST	/api/cases/:id/report	Generate report.
GET	/api/reports/:id	Retrieve report metadata/download.
GET	/api/analytics/summary	Dashboard metrics.
GET	/api/audit/cases/:id	Retrieve case audit trail.

AI service API

POST /predict → accepts one image and returns quality result, detection list, counts, WBC classifications, confidence values and review-priority fields. GET /health → service health. The exact JSON schema must be versioned and stored in the repository so frontend/backend changes do not silently break the inference service.

13. Web Application Requirements

Screen	Main contents
Login	Email/password, role-based routing, basic session state.
Laboratory Dashboard	Pending cases, processing cases, review queue, completed cases, basic statistics.
Create Case	Case/sample ID, subject code, optional minimum metadata, assigned reviewer.
Case Detail	Images, status, AI results, reviewer status, previous reports, audit trail.
Image Upload	Drag/drop or file picker, multiple image support, preview, remove/reorder.
Quality Review	Score, reasons, accept/re-upload recommendation.
Analysis Results	Cell counts, annotated image, WBC subtype counts, confidence distribution.
Cell Review	Cell gallery, filters by class/confidence/priority, accept/reclassify/unknown, comments.
Report Preview	Verified counts, reviewed findings, disclaimer, approval status, PDF generation.
History	Search previous cases, reports and analysis summaries.
Admin	User management and system statistics; keep simple for MVP.

UI principle

The interface should make the AI's status obvious: **AI suggestion**, **confidence**, **review required**, and **final verified result** must never be visually conflated.

14. Laboratory Workflow and Human Verification

The system assumes that the laboratory technician is responsible for capturing or obtaining the microscopic image. CellInsight does not physically recapture an image unless a camera integration is later added. In the two-month MVP, upload mode is the guaranteed feature; camera capture is optional.

Step	Actor	System action
1. Create case	Technician	Create case/sample record.
2. Add smear images	Technician	Upload one or more microscope images.
3. Quality check	System	Assess focus/brightness/contrast/resolution and return status.
4. Recapture/re-upload if needed	Technician	If image is poor, adjust microscope/capture setup and upload another image.
5. AI analysis	System	Run YOLO + WBC classifier on accepted images.
6. Review queue	System	Prioritise low-confidence/abnormal-looking cells.
7. Verification	Technician/Doctor	Accept, reclassify or mark unknown; add comments if needed.
8. Approval	Doctor/authorised reviewer	Approve final verified case/report.
9. Archive	System	Store images, AI output, reviewer actions and report.

Important positioning

The AI output is a pre-analysis/assistance layer. The application must not claim that a blast-like flag equals leukemia or that a simple WBC count threshold equals a clinical risk diagnosis.

15. Reporting and Digital Records

The PDF is no longer the product. It is the final output of the verified workflow.

Report section	Content
Case header	Case ID, sample ID, date/time, model version, reviewer status.
Image summary	Number of images/fields, quality status and rejected-field count.
Cell summary	RBC, WBC, platelet counts and WBC subtype counts.
Review summary	Number accepted, modified, unknown and flagged.
Abnormal/uncertain section	Cells requiring review and final verified status; avoid diagnostic claims.
Annotated evidence	Selected annotated images/cell crops where useful.
Audit information	Reviewer, approval time, report version.
Disclaimer	Research prototype; not a standalone diagnostic system; final interpretation remains with qualified professionals.

Archive requirements

- Every report must have a version number.
- A corrected report must not silently overwrite an earlier approved report.
- Original images should remain linked to the case.
- Reviewer actions should be retrievable.
- Model version used for an analysis must be stored.

16. Security, Privacy and Safety

Area	MVP requirement
Authentication	Password hashing; no plaintext passwords.
Authorisation	Role-based access checks on every protected API.
Input validation	Validate file type, file size, IDs and API payloads.
File safety	Store uploads outside executable code paths; generate safe filenames.
Audit	Record case creation, review changes, approval and report generation.
Privacy	Use synthetic/de-identified data for development and demo whenever possible.
Patient identifiers	Minimise stored personally identifying data; use case/subject codes for prototype datasets.
Medical safety	No autonomous diagnosis, treatment recommendation or false claims of clinical validation.
Error handling	Show safe user messages; keep technical logs server-side.

Clinical boundary

This project is a student research prototype. It should be demonstrated as an AI-assisted review and workflow system, not as a certified medical device. Any real-world clinical deployment would require appropriate validation, governance, privacy controls, laboratory verification and regulatory assessment.

17. Evaluation and Success Metrics

The final project should measure both **AI performance** and **workflow usefulness**. This directly addresses the criticism that simply connecting existing models is insufficient.

Category	Metrics	Purpose
Detection	Precision, Recall, F1, mAP	Measure cell-detection quality.
Classification	Accuracy, macro F1, per-class precision/recall, confusion matrix	Measure WBC classification.
Image quality	Quality score distribution; accepted vs rejected images; relationship to model confidence	Validate the quality gate.
Human review	AI-human agreement, correction rate, unknown rate	Measure usefulness of AI preclassification.
Efficiency	AI processing time, reviewer time/case, total workflow time	Test whether the system reduces repetitive effort.
Reliability	API error rate, failed analyses, invalid files, repeatability	Assess software reliability.
Usability	Task completion observations / short user questionnaire	Assess practical interface quality.

Suggested comparison experiment

- **Manual:** reviewer examines the selected dataset without AI assistance.
- **AI-assisted:** reviewer receives AI preclassification and only verifies/corrects it.
- Measure time per case, agreement/accuracy and number of corrections.

18. Testing Strategy

Level	Examples
Unit tests	Quality-score functions, confidence logic, review-priority calculation, API validation, database helpers.
AI service tests	Known image → expected JSON structure; missing model; corrupt image; oversized image; low-quality image.
Backend integration	Upload → analysis → save → review → approve → report.
Frontend tests	Login routing, upload states, loading/error states, filters, review actions.
Security tests	Unauthorised role attempts, invalid tokens, unsafe filenames, invalid file types.
Performance tests	Multiple images in one case; repeated inference; concurrent API requests at small prototype scale.
User acceptance	Technician scenario: create case → upload → review → approve → report.

Definition of done for each module

A module is not considered complete merely because its code runs. It must have: working integration, basic error handling, documented inputs/outputs, at least one test, and a demonstrable result in the application.

19. Deployment and Development Environment

Component	Development	Demo/deployment
Frontend	React + Vite, local browser	Vercel or local Docker/static server if desired.
Backend	Node.js + Express	Local server or free-tier host where practical.
AI service	Python + FastAPI/Uvicorn	Local machine or suitable free/low-cost host; CPU inference preferred for demo.
Database	MongoDB local / free tier	MongoDB deployment if needed; otherwise local for final demo.
Training	Google Colab free tier	Models exported as versioned artefacts.
Source control	Git/GitHub	Tagged release for final demo.

Recommended MVP deployment

For the final college demonstration, local deployment is acceptable and safer: run React, Express, FastAPI and MongoDB on a development laptop, with the trained model files available locally. A cloud deployment is optional and should not be allowed to delay the core project.

20. Eight-Week Implementation Plan

Week	Primary work	Exit condition
1	Repository audit; final taxonomy; architecture; database/API design; baseline model evaluation.	Architecture frozen; existing models run reproducibly.
2	Authentication, roles, case management, MongoDB collections, base dashboard.	User can create/search/update cases.
3	Image upload, multi-image support, metadata, OpenCV quality gate.	Image quality workflow works end-to-end.
4	FastAPI inference integration; YOLO + classifier; structured cell results.	Accepted image produces stored AI analysis.
5	Cell review UI; confidence; review-priority logic; accept/reclassify/unknown; audit events.	Human-in-the-loop workflow works.
6	Verified report, archive/history, dashboard analytics, optional email.	Case can be approved, archived and reported.
7	Model/workflow evaluation; testing; performance measurements; bug fixing.	Evaluation tables and test evidence complete.
8	Deployment/demo polish; documentation; screenshots; PPT; final PRD alignment.	Stable final demo and submission package.

Scope rule for the two-month window

If time becomes tight, cut optional camera capture, email automation, advanced explainability and cloud deployment before cutting human verification, case management, quality gate, AI integration, database persistence or evaluation.

21. Team Work Allocation

Workstream	Primary owner	Secondary support
AI/model evaluation, quality scoring, inference service	Member A	Member B
Backend, MongoDB, authentication, APIs, audit	Member B	Member C
Frontend, dashboard, review UI, report UI	Member C	Member B
Integration/testing/documentation	All	All

Parallel development rule

The three workstreams should use stable contracts early. The AI service must expose a versioned JSON response; the backend must expose documented REST endpoints; the frontend should consume those contracts rather than directly reading model files. This prevents the team from waiting for one person to finish before others can work.

Git workflow

- main → stable/demo branch.
- feature/ → individual work.
- Pull request before merging.
- Tag model versions and final demo release.
- Keep a small CHANGELOG for API/model schema changes.

22. Coding-Agent Instructions

The following rules are intended to be copied into the coding agent's project instructions.

- Treat this PRD as the source of truth for scope unless a human explicitly approves a change.
- Do not rewrite the existing YOLO/EfficientNet models unless evaluation shows that a change is necessary.
- Do not add disease diagnosis, treatment recommendations, patient portal, or large hospital-management features.
- Do not hardcode clinical thresholds such as 'WBC > 20 = normal/risk'.
- Never claim that a blast-like model prediction is a diagnosis of leukemia.
- Every AI result must carry a model version and confidence where available.
- Keep original AI predictions immutable; reviewer corrections must be stored separately.
- Implement backend validation and role checks; never rely only on frontend restrictions.
- Return consistent API error formats.
- Write tests for new business logic and critical AI-service functions.
- Keep image files separate from database metadata; do not put large image binaries into MongoDB unless there is a deliberate reason.
- Prefer simple, free/open-source solutions. Avoid paid infrastructure unless explicitly approved.
- Do not add optional features until all Must requirements are working.
- Keep UI labels medically cautious: 'AI suggestion', 'review required', 'blast-like/abnormal flag', not 'diagnosis'.
- Before changing the database schema or API contract, update the schema documentation and affected consumers.

Suggested agent execution order

- Inspect repository and map existing files to Modules 1–5.
- Run the current system and establish a baseline test.
- Freeze API/data contracts.
- Implement database and case management.
- Integrate image-quality service.
- Integrate current AI models through FastAPI.
- Implement human review and audit trail.
- Implement report/archive.
- Add evaluation scripts and tests.
- Only then polish UI and optional integrations.

23. Acceptance Criteria

ID	Acceptance criterion
AC-01	A technician can log in and create a new case.
AC-02	A case can contain multiple smear images.
AC-03	Each uploaded image receives a quality status and score/reasons.
AC-04	Poor-quality images are not silently processed as if they were valid; the user is asked to re-upload/recapture.
AC-05	An accepted image reaches the FastAPI AI service and returns structured detections.
AC-06	WBC detections receive subtype predictions and confidence values where supported.
AC-07	The UI distinguishes AI predictions from final verified labels.
AC-08	A reviewer can accept, reclassify or mark a cell unknown.
AC-09	The system stores original AI prediction and reviewer correction separately.
AC-10	A case can be approved and a PDF report generated from verified data.
AC-11	The case and report can be retrieved later from the laboratory dashboard.
AC-12	Model version and audit information are preserved.
AC-13	The system produces evaluation metrics for detection/classification and workflow metrics for review.
AC-14	The system does not display unsupported clinical diagnosis/risk claims.
AC-15	A complete demo can be executed locally without paid infrastructure.

24. Risks and Mitigations

Risk	Impact	Mitigation
Insufficient/poor dataset	High	Use public datasets for development; use a clearly separated validation set; document dataset limitations.
Current WBC taxonomy is biologically inconsistent	High	Review labels before retraining/evaluation; separate NRBC/erythroblast from WBC classes.
Model accuracy is lower than expected	High	Keep model as assistance only; use confidence/unknown class; focus evaluation on transparent limitations.
Scope grows beyond 8 weeks	High	Must/Should/Could prioritisation; cut optional camera/cloud/email before core workflow.
Camera integration unavailable	Medium	Make upload mode the guaranteed MVP; camera capture optional.
Medical overclaiming	High	Use research-prototype language and human verification; no autonomous diagnosis.
API/schema mismatch between team members	Medium	Freeze JSON contracts and version them.
Storage becomes messy	Medium	Separate metadata and file paths; use case IDs and deterministic folder structure.
Security/privacy issues with real patient data	High	Use synthetic/de-identified demo data and minimise PII.
Training consumes too much time	Medium	Use Colab/free GPU only for targeted experiments; prioritise integration and evaluation.

25. Future Scope

- Microscope-camera capture and guided live acquisition.
- Automatic multi-field scanning and slide coverage.
- Expanded WBC and RBC morphology classes.
- Additional abnormal-cell models with stronger external validation.
- Explainable-AI heatmaps and morphology feature summaries.
- LIS/HIS integration.
- Cloud-based multi-laboratory review and remote consultation.
- Model retraining from verified reviewer corrections.
- Advanced quality-control dashboards.
- Formal clinical validation and regulatory pathway assessment before any real clinical use.

Important boundary

Future scope is intentionally separated from the two-month MVP. The final project should demonstrate a coherent, complete core system rather than many partially implemented features.

26. References and Source Notes

Project-provided sources

- CellInsight project synopsis, Academic Year 2026–27, supplied by the project team. It defines the approved title, problem statement, objectives, methodology, scope, tools and datasets.
- Project Review–I document, Department of Information Technology, Academic Year 2026–27, supplied by the project team. It records review criteria and the current architecture/workflow presentation.
- Shared implementation snapshot supplied in the conversation. It describes the current three-service implementation: FastAPI inference, Express backend and React/Vite frontend.

External research / benchmark sources

- Kratz A. et al. (ICSH). Digital morphology analyzers in hematology: ICSH review and recommendations. International Journal of Laboratory Hematology, 2019. PMID 31046197. The review supports AI preclassification, digital archiving, QA, consultation and the continued role of skilled morphologists.
- CellaVision. Peripheral Blood Application. Product documentation. Describes AI pre-classification followed by Medical Technologist review/verification, reclassification, comments, comparison and archival.
- CellaVision. Review Software. Product documentation. Describes review/verification workflows, collaboration and comparison with previous results.
- Briggs C. et al. ICSH guidelines for evaluation of blood cell analysers including digital image analysis. International Journal of Laboratory Hematology, 2014. PMID 24666725.
- Development of a Low-cost Smartphone-connected Digital Microscope. Journal article indexed by PubMed, PMID 32766118. Supports feasibility of low-cost smartphone-connected microscopy approaches.

Source URLs used during PRD research

<https://pubmed.ncbi.nlm.nih.gov/31046197/>

<https://www.cellavision.com/products/software/cellavisionr-peripheral-blood-application>

<https://www.cellavision.com/products/software/cellavisionr-review-software>

<https://pubmed.ncbi.nlm.nih.gov/24666725/>

<https://pubmed.ncbi.nlm.nih.gov/32766118/>

27. One-Page Build Summary

Area	Final decision
Project identity	CellInsight — AI-Assisted Digital Peripheral Blood Smear Review Platform
Core users	Laboratory technician, doctor/pathologist, administrator
Core AI	YOLOv8 + EfficientNet/Keras WBC classifier
New intelligence	Image quality scoring, confidence handling, review prioritisation, correction tracking
Core software	React + Node/Express + FastAPI + MongoDB
Core workflow	Case → images → quality gate → AI → review → approval → archive → report
Clinical boundary	Assistance/review support only; no autonomous diagnosis
MVP hardware	Existing microscope + existing camera/smartphone if available; upload mode guaranteed
Five implementation modules	Data/model → AI service → backend/database → web app → testing/deployment
Evaluation	Model metrics + AI-human agreement + correction rate + review time + image rejection
Time	8 weeks
First thing to build	Repository audit + architecture/API/data contracts
Last thing to polish	Optional camera/email/cloud features

Final product definition

A laboratory-side web system that uses AI to pre-analyse peripheral blood-smear images, highlights uncertain cells for human verification, stores the complete case history and produces a verified report. The project is valuable because the AI models become part of a usable, measurable and traceable workflow.

END OF PRD